

How to Prevent Cookies from Being Hijacked

Every time you log in to a website, you leave a footprint in the form of cookies. These can be used to gain unauthorised access to the information on your system. Let's take a look at how AES 128 can be used to prevent cookie hijacking.



Image Source: By jcomp on Freepik

Cookies store small blocks of data created by a web server to help the site remember information about your visit. This is helpful for session management during logins and auto-filled form fields, user personalisation (where cookies retain information on user preferences and themes), and for tracking and analysing your web browsing pattern to recommend personalised ads, etc.

You may be wondering how long these cookies retain information. Well, there are different types of cookies. Session cookies store information only till you exit the browser, first-party cookies are stored directly on your computer and help with auto logins and more, secure cookies prevent unauthorised entry, and zombie cookies are present even when they have been deleted or the browser is exited.

The presence of these cookies can lead to unauthorised access to information or services in your computer system. If you look at your normal internet activity, you will find you visit quite a few websites every day. Since we enter sensitive information such as passwords, date of birth, bank account details, etc, when we visit these websites, there are chances this information may fall in the hands of an attacker who is trying to explore the vulnerabilities in the cookies. The attacker may steal the cookie using a fake login or link, put the cookie in the browser, and may fake your identity in the browser. This is known as cookie hijacking or session hijacking because it includes the exploitation of the session key (symmetric key for encrypting data). Common methods used by attackers for